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CSC 112 – BNO1
ASSIGNMENT

1. Prove to be a RL using pumping lemma (show proof) and give equivalent automaton if it's a regular language. (10 pts. Each)

   1. L = {0^k1^k | k >= 1}
      a. PROOF
         i. Assume that L is a regular language;
         ii. And pumping length = P
         iii. S = 0^P1^P, where S is divided into 3 parts S = x y z
         iv. If P (pumping length) is 6, then S = 000000111111
         v. Dividing S into x y z
            **Note: highlighted texts ## is x, ## is y, ## is z.*
            1. Case 1: y is in 0 part
               a. S = 00 0000 111111
                  $xy^iz \Rightarrow xy^2z$
               b. x = 00
               c. y = 0000 0000
               d. z = 111111
               e. 0's = 10
                  1's = 6
               f. 10 $\neq$ 6 $\rightarrow S \notin L$
            2. Case 2: y is in 1 part
               a. 000000 1111 11
                  $xy^iz \Rightarrow xy^2z$
               b. x = 000000
               c. y = 1111 1111
               d. z = 11
               e. 0's = 6
                  1's = 10
               f. 6 $\neq$ 10 $\rightarrow S \notin L$
            3. Case 3: y is in the '0' and '1' part
               a. 0000 0011 1111 x = 00
                  $xy^iz \Rightarrow xy^2z$
               b. x = 0000
               c. y = 0011 0011
               d. z = 1111
               e. S does not follow 0's1's pattern. Therefore, $S \notin L$.
         vi. $|xy| \leq P$
            1. Case 1: P = 6, x = 2, y = 4
               a. $|6| \leq 6. True$
            2. Case 2: P = 6, x = 6, y = 4
               a. $|10| \leq 6. False$
            3. Case 3: P = 6, x = 4, y = 4
               a. $|8| \leq 6. False$
         vii. The following conditions:
            1. $xy^iz \in L$ *for every* $i \geq 0$ *is False.*
            2. $|y| > 0$ *is True.*
            3. $|xy| \leq P$ *is False.*

         $\therefore$ *S cannot be pumped == contradicts L is a Regular Language.*

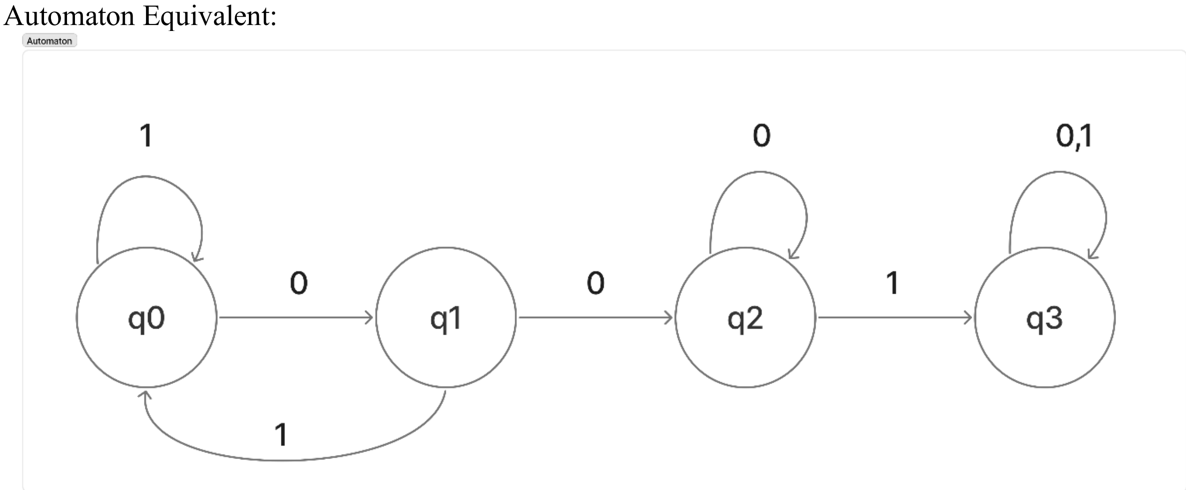
2. L = {w | w contains the substring 001}
    a. PROOF
        i. Assume that L is a regular language;
        ii. And pumping length = P
        iii. String S = $0^P 0\ 0\ 1\ 1^P$, clearly S contains the substring “001”, so $S \notin L\ and\ |S|\ \geq P$
        iv. If P (pumping length) is 6, then S = 0^6 001 0^6 = 000000 001 000000
        This means that the x and y are located within the first P symbols of S = (0^P 0 0 1 1^P).
        v. Dividing S into x y z
            1. Case 1: y is in the leading 0 part
                a. S = 00 0000 001000000
                $xy^iz \Rightarrow xy^2z$
                b. x = 00
                c. y = 0000 0000
                d. z = 001000000
                e. $Specific\ 001\ substring\ remains\ intact.\ Therfore, S \notin L.$
            2. Case 2: y is in the 001 part
                a. The “001” part begins after position P. But the condition $|xy| \leq P$ says y ends at or before position P. Hence, y cannot be in the “001” portion.
                b. Case 2 is false (impossible)
            3. Case 3: y is in the trailing 0s (after the 001)
                a. Same reasoning as case 2 meaning y appears before the “001” substring and cannot extend beyond it. So, y cannot be in the trailing 0’s.
                b. Case 3 is false (impossible)

            Only Case 1 is valid and satisfies the Pumping Lemma for this language.

        vi. The following conditions:
            1. $xy^iz\ \in L\ for\ every\ i \geq 0\ is\ True\ shown\ in\ Case\ 1.$
            2. $|y|\ > 0\ is\ True.$
            3. $|xy| \leq P\ is\ True.$

$\therefore S\ can\ be\ pumped\ (the\ pumped\ strings\ stay\ in\ L).$
$Threfore, L\ satisfies\ the\ Pumping\ Lemma\ and\ is\ a\ Regular\ Language$

Automaton Equivalent:



Provide a CFG for the following expressions, and prove that each expression is in L(G) by giving a derivation and the corresponding parse tree. (10 pts. each)

Context-Free Grammar:

G = {V, Σ, R, S}
V = {E, T, F, N}
Σ = {+, -, *, /} ∪ {N | N ∈ ℝ}
S = E
R:

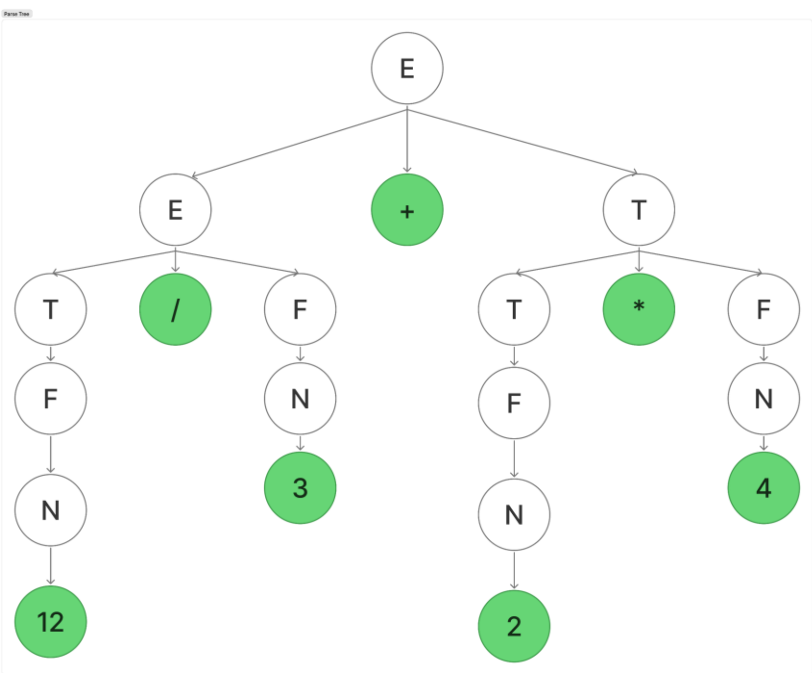
$E \rightarrow E - T \mid E + T \mid T$
$T \rightarrow T * F \mid T / F \mid F$
$F \rightarrow E \mid N$

1. 12 / 3 + 2 * 4

Leftmost Derivation of 12 / 3 + 2 * 4:

1.	$E \rightarrow E + T$	{as $E \rightarrow E + T$}
2.	$E \rightarrow T + T$	{as $E \rightarrow T$}
3.	$E \rightarrow T/F + T$	{as $T \rightarrow T/F$}
4.	$E \rightarrow F/F + T$	{as $T \rightarrow F$}
5.	$E \rightarrow N/F + T$	{as $F \rightarrow N$}
6.	$E \rightarrow 12/F + T$	{as $N \rightarrow 12$}
7.	$E \rightarrow 12/N + T$	{as $F \rightarrow N$}
8.	$E \rightarrow 12/3 + T$	{as $N \rightarrow 3$}
9.	$E \rightarrow 12/3 + T * F$	{as $T \rightarrow T * F$}
10.	$E \rightarrow 12/3 + F * F$	{as $T \rightarrow F$}
11.	$E \rightarrow 12/3 + N * F$	{as $F \rightarrow N$}
12.	$E \rightarrow 12/3 + 2 * F$	{as $N \rightarrow 2$}
13.	$E \rightarrow 12/3 + 2 * N$	{as $F \rightarrow N$}
14.	$E \rightarrow 12/3 + 2 * 4$	{as $N \rightarrow 4$; End}

LEFT PARSE TREE:

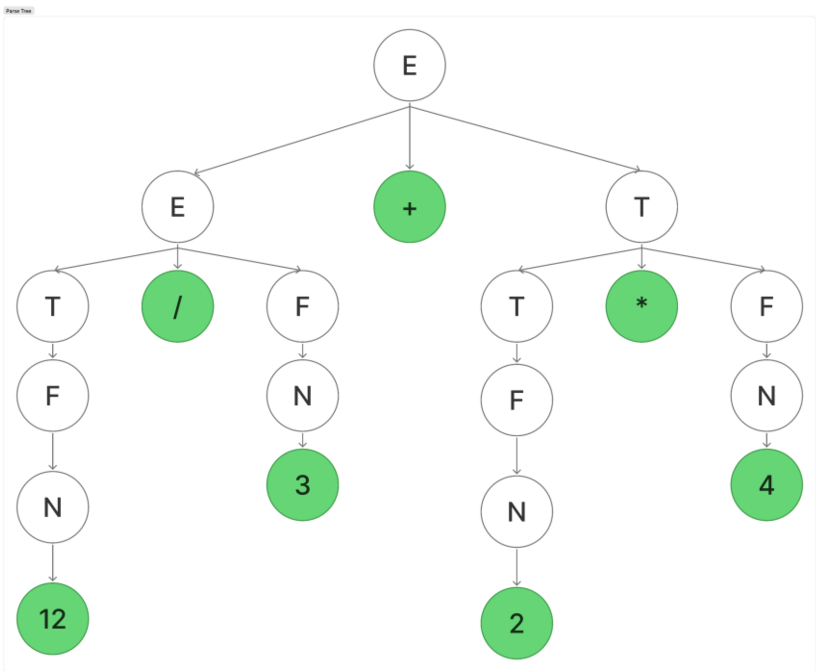


2.

Rightmost Derivation of 12 / 3 + 2 * 4:

1.	$E \rightarrow E + T$	{as $E \rightarrow E + T$}
2.	$E \rightarrow E + T * F$	{as $T \rightarrow T * F$}
3.	$E \rightarrow E + T * N$	{as $F \rightarrow N$}
4.	$E \rightarrow E + T * 4$	{as $N \rightarrow 4$}
5.	$E \rightarrow E + F * 4$	{as $T \rightarrow F$}
6.	$E \rightarrow E + N * 4$	{as $F \rightarrow N$}
7.	$E \rightarrow E + 2 * 4$	{as $N \rightarrow 2$}
8.	$E \rightarrow T + 2 * 4$	{as $E \rightarrow T$}
9.	$E \rightarrow T / F + 2 * 4$	{as $T \rightarrow T/F$}
10.	$E \rightarrow T / N + 2 * 4$	{as $F \rightarrow N$}
11.	$E \rightarrow T / 3 + 2 * 4$	{as $N \rightarrow 3$}
12.	$E \rightarrow F / 3 + 2 * 4$	{as $T \rightarrow F$}
13.	$E \rightarrow N / 3 + 2 * 4$	{as $F \rightarrow N$}
14.	$E \rightarrow 12 / 3 + 2 * 4$	{as $N \rightarrow 12$}

RIGHT PARSE TREE:

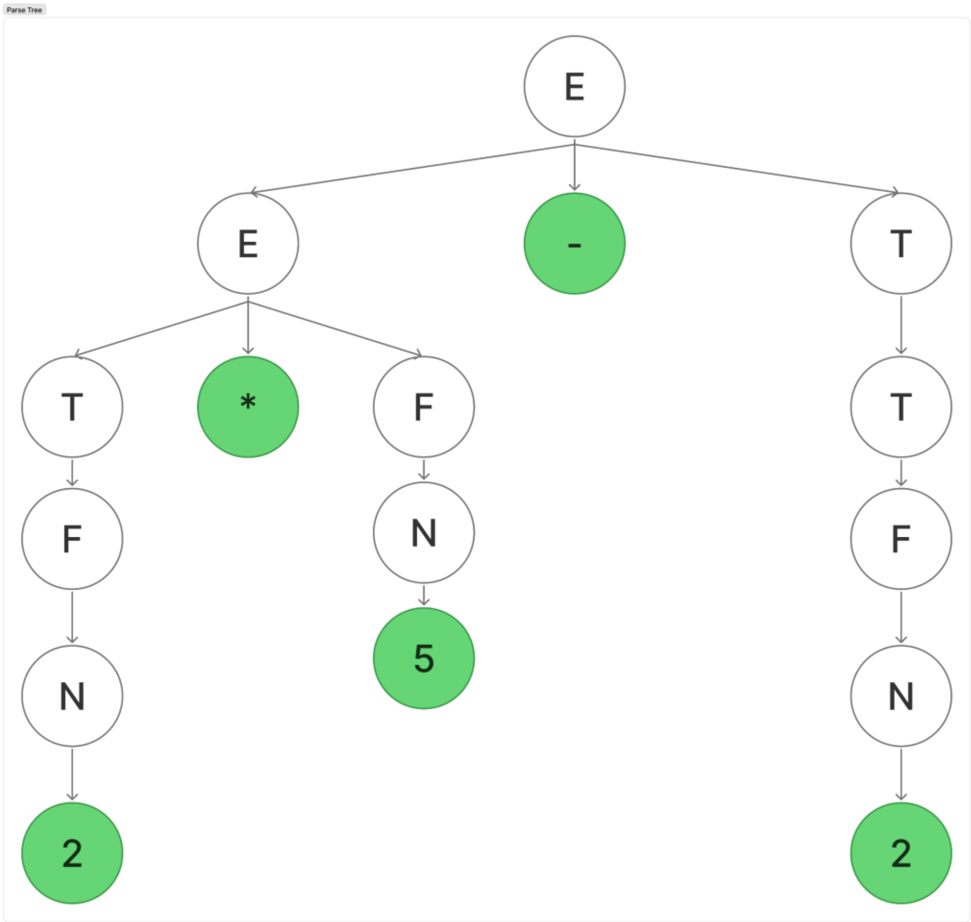


2. 2 * 5 - 2

Leftmost Derivation of 2 * 5 – 2:

1. $E \rightarrow T$	{as $E \rightarrow T$}
2. $E \rightarrow T - T$	{as $T \rightarrow T - T$}
3. $E \rightarrow T * F - T$	{as $T \rightarrow T * F$}
4. $E \rightarrow F * F - T$	{as $T \rightarrow F$}
5. $E \rightarrow N * F - T$	{as $F \rightarrow N$}
6. $E \rightarrow 2 * F - T$	{as $N \rightarrow 2$}
7. $E \rightarrow 2 * N - T$	{as $F \rightarrow N$}
8. $E \rightarrow 2 * 5 - T$	{as $N \rightarrow 5$}
9. $E \rightarrow 2 * 5 - F$	{as $T \rightarrow F$}
10. $E \rightarrow 2 * 5 - N$	{as $F \rightarrow N$}
11. $E \rightarrow 2 * 5 - 2$	{as $N \rightarrow 5$; End}

LEFT PARSE TREE:



Rightmost Derivation of 2 * 5 – 2:

1. $E \rightarrow T$	{as $E \rightarrow T$}
2. $E \rightarrow T - T$	{as $T \rightarrow T - T$}
3. $E \rightarrow T - F$	{as $T \rightarrow F$}
4. $E \rightarrow T - N$	{as $F \rightarrow N$}
5. $E \rightarrow T - 2$	{as $N \rightarrow 2$}
6. $E \rightarrow T * F - 2$	{as $T \rightarrow T * F$}
7. $E \rightarrow T * N - 2$	{as $F \rightarrow N$}
8. $E \rightarrow T * 5 - 2$	{as $N \rightarrow 5$}
9. $E \rightarrow F * 5 - 2$	{as $T \rightarrow F$}
10. $E \rightarrow N * 5 - 2$	{as $F \rightarrow N$}
11. $E \rightarrow 2 * 5 - 2$	{as $N \rightarrow 2$; End}

RIGHT PARSE TREE: